Week 5 — Geometry and Elimination

Linear Algebra I, TT 2026

Based on Strang, *Introduction to Linear Algebra*, 4th ed., §§2.1–2.3 and *Linear Algebra and Its Applications*, 4th ed., §§1.2–1.4.

80 minutes of lecture; 90 minutes of guided problems.

Timing

Review of rank and inverses: 10 min.

Row and column pictures: 20 min.

Elimination and back substitution: 25 min.

Elimination matrices E and permutation matrices P: 20 min.

Summary and questions: 5 min.

Review: rank and inverse

$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has rank 1 and no inverse.

$A\mathbf{x} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ has infinitely many solutions, whereas $A\mathbf{x} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$ has none.

For a square $n \times n$ matrix, n pivots mean invertibility and a unique solution for each $\mathbf{b}$.

From last week to this week

Gauss–Jordan continues to reduced echelon form. Gaussian elimination stops at an upper triangular U and uses back substitution.

Both act on the full augmented matrix $[A \mid \mathbf{b}]$.

Row picture: two lines

$$2x - y = 1, \qquad x + y = 5.$$

$$\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}.$$

The two lines intersect at $(x, y) = (2, 3)$. Each row is one geometric constraint.

Column picture: two vectors

$$x \begin{bmatrix} 2 \\ 1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}.$$

$$2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}.$$

Strang emphasizes: $A\mathbf{x}$ is a linear combination of the columns of A.

Rows versus columns

Rows: two lines in 2D, three planes in 3D; seek their common intersection.

Columns: choose coefficients $x_1, \ldots, x_n$ so that $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$.

Both are the same equation $A\mathbf{x} = \mathbf{b}$.

Three planes

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}.$$

The three equations describe planes intersecting at $(u, v, w) = (1, 1, 2)$.

Three-dimensional column picture

$$u\begin{bmatrix}2\\4\\-2\end{bmatrix}+v\begin{bmatrix}1\\-6\\7\end{bmatrix}+w\begin{bmatrix}1\\0\\2\end{bmatrix}=\begin{bmatrix}5\\-2\\9\end{bmatrix}.$$

The combination with weights $(1,1,2)$ produces $\mathbf{b}$.

Three possible outcomes

One solution: independent constraints.

Infinitely many: $x + y = 2$, $2x + 2y = 4$.

No solution: $x + y = 2$, $2x + 2y = 5$.

Elimination identifies $0 = 0$ or $0 = c \neq 0$.

Pivot and multiplier

To eliminate entry a_{ij} below a nonzero pivot a_{jj}, use $R_i \leftarrow R_i - \ell_{ij} R_j$ where $\ell_{ij} = a_{ij}/a_{jj}$.

Forward elimination produces an upper triangular matrix; back substitution then solves the system.

A complete 3 by 3 example

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 2 & 9 \end{array}\right].$$

Pivot 2, multipliers $\ell_{21} = 2$, $\ell_{31} = -1$.

Use $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 + R_1$.

First stage of elimination

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 8 & 3 & 14 \end{array}\right].$$

Second pivot -8, multiplier $\ell_{32} = -1$.

Use $R_3 \leftarrow R_3 + R_2$.

Upper triangular form

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 0 & 1 & 2 \end{array}\right].$$

Pivots $2, -8, 1$; $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix}$.

Back substitution

$w = 2$, $-8v - 2w = -12 \Rightarrow v = 1$, $2u + v + w = 5 \Rightarrow u = 1$.

$$\mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

Verify in the original $A\mathbf{x} = \mathbf{b}$.

Why elimination preserves solutions

Row replacement $R_2 \leftarrow R_2 - 2R_1$ is reversible: add $2R_1$ back.

Apply the same operation to the right side; otherwise the transformed equations are not equivalent.

Elementary matrix E_{21}

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$E_{21} \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ -2 & 7 & 2 \end{bmatrix}.$$

Left multiplication performs a row operation.

The right side also transforms

$$E_{21}\begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix} = \begin{bmatrix} 5 \\ -12 \\ 9 \end{bmatrix}.$$

Therefore $A\mathbf{x} = \mathbf{b}$ becomes $(E_{21}A)\mathbf{x} = E_{21}\mathbf{b}$.

Other elimination matrices

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \; E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

$E_{32}E_{31}E_{21}A = U.$

Rightmost factor acts first; matrix multiplication order matters.

Undo an elimination step

$E = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$ and $E^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$.

$E^{-1}E = I$.

Subtracting three times row 1 is undone by adding three times row 1.

Zero pivot: row exchange

$$\left[\begin{array}{cc|c} 0 & 2 & 4 \\ 3 & 1 & 7 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 3 & 1 & 7 \\ 0 & 2 & 4 \end{array}\right].$$

Swap rows; $y = 2$, $x = 5/3$. A zero pivot does not necessarily imply singularity.

Permutation matrix P

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \; P \begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}, \; P \begin{bmatrix} 4 \\ 7 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}.$$

The equivalent system is $PA\mathbf{x} = P\mathbf{b}$.

Properties of P

One 1 in each row and column, zeros elsewhere.

$P^{-1} = P^T$; for a single swap $P^2 = I$.

$$P_{13} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

Left multiplication exchanges equations, not unknowns.

Temporary versus permanent failure

Temporary: zero pivot with nonzero entry below; swap rows.

Permanent: no available pivot in the required column; singular systems may have no solutions or infinitely many.

Next week: LU and LDU

Without row exchanges, $A = LU$: U is upper triangular; L stores elimination multipliers.

With row exchanges, $PA = LU$.

Week 6 constructs and uses these factorizations.

Summary and reading

Rows = intersections; columns = linear combinations.

Elimination = zero entries below pivots; back substitution solves.

E implements row replacement; P implements row swaps.

Reading: Strang, *Introduction to Linear Algebra*, 4th ed., §§2.1–2.3.